

PRODUCT REQUIREMENTS DOCUMENT

Tuition Management System (TMS)

Multi-Tenant · Web & Mobile · Nepal

|                                    |                                     |                                             |                                                       |
|------------------------------------|-------------------------------------|---------------------------------------------|-------------------------------------------------------|
| <div>Version</div> <div>v2.0</div> | <div>Date</div> <div>May 2026</div> | <div>Status</div> <div>Conceptual PRD</div> | <div>Stack</div> <div>React · Node · PostgreSQL</div> |
|------------------------------------|-------------------------------------|---------------------------------------------|-------------------------------------------------------|

## Document Changelog

| Version | Date        | Author   | Changes                                                                                                                                                                                                                                                                                         |
|---------|-------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0    | May 2026    | Internal | Initial PRD — core platform, geo-attendance, billing, leave, appointments, petty cash, resource logging.                                                                                                                                                                                        |
| v2.0    | 10 May 2026 | Internal | Added: Social Media Integration, Certificate Generation, Academic Calendar, Personalized Classes, Financial Estimation & AI Suggestions, Yearly Financial Tracking, HR Management (full), Staff Performance Scoring, Student Performance Tracking, Canteen & Vehicle Tracking (Phase 2 scoped). |

# 1. Executive Summary

The Tuition Management System (TMS) is a cloud-based, multi-tenant SaaS platform for private tuition institutions in Nepal. It serves three core pillars:

|                                                                                                                                |                                                                                                                    |                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Institution Management</b><br>Classes, attendance, teachers, students, timetables, academic performance, homework, results. | <b>ERP &amp; Finance</b><br>Fee billing, Nepal Pay, VAT/PAN invoicing, payroll, petty cash, expense tracking, P&L. | <b>Communication Hub</b><br>Push + SMS notifications, parent-initiated messaging, appointment booking, social media management. |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|

TMS covers UKG to Grade 12, music classes, personalized classes, and short/long-term courses. It supports multiple branches per tenant, RBAC, geo-fenced teacher attendance, automated workflows, AI-assisted financial estimation, HR management, certificate generation, and social media scheduling.

## 2. Goals & Objectives

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### 2.1 Business Goals

- Provide a single platform for all academic, administrative, HR, and financial operations of a tuition institution.
- Enable multi-tenant deployment with full data isolation between institutions.
- Support Nepal-specific requirements: Nepal Pay, VAT/PAN, SMS gateway.
- Automate repetitive tasks — fee reminders, homework, attendance reports, salary calculations, certificate generation.
- Deliver mobile-first experience for teachers, students, and parents.
- Enable institutions to build their social media presence directly from TMS.

### 2.2 Product Goals

- Modular architecture — every feature configurable by Tenant Admin within Super Admin policies.
- RBAC-driven — all permissions flow from role definitions.
- Zero-assumption design — all thresholds and rules are admin-configurable.
- AI-assisted financial forecasting and expense suggestions from Phase 2 onwards.

## 3. Scope

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### 3.1 Phase 1 — Core Foundation

- Multi-tenant setup · RBAC · Student & teacher enrollment · Regular class timetable (UKG–12)
- Full geo-attendance system for teachers
- Student attendance (teacher-managed)
- Fee billing engine · Nepal Pay QR · VAT/PAN invoicing · Webhooks
- Student fee blocking and admin override
- Smart admin dashboard · Digital student ID · Lifetime record
- Leave and early out system (staff + student) · Student emergency departure
- Push + SMS notification foundation

### 3.2 Phase 2 — Academic, Operations & Intelligence

- Homework management + auto-distribution · Results & performance tracking
- Teacher verification (mandatory daily logout gate)
- Communication hub · Appointment booking · Resource & infrastructure logging
- Certificate generation (all types, PDF, admin-issued)
- Academic calendar (tenant + branch level, visible to students/parents)
- Social media post scheduling (Facebook, Instagram, TikTok, LinkedIn)
- Personalized classes (one-on-one / small group, custom fee and rules)
- Student performance tracking (upgrade/downgrade vs time + class average)
- HR management (records, documents, appraisals, exit flow)
- Staff performance scoring (continuous, auto-updated)
- AI-driven financial estimation and expense suggestions
- Yearly financial tracking · Calendar payment view
- Canteen management system
- Vehicle tracking system
- Report generation + PDF/Excel export · Teacher rating & feedback

### 3.3 Phase 3 — ERP & Advanced

- Full expense management · Payroll auto-calculation · Petty cash approval engine
- P&L dashboard · Double-entry ledger export
- Music class module · Short/long-term course engine · Refund management
- Full cron automation suite · Task escalation engine

### 3.4 Out of Scope

- Library management
- Virtual/online class sessions (planned future version)
- Course video hosting (planned future version)

## 4. Users & Roles

The system uses strict RBAC. All permissions are defined by role and scoped to the hierarchy level of the authenticated user.

| Role                     | Scope            | Key Permissions                                                                                                                                                                                                                |
|--------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Super Admin</b>       | Platform         | Full access to all tenants. Audit/support login into any tenant. Manages onboarding, platform config, default policies.                                                                                                        |
| <b>Tenant Admin</b>      | Institution      | All centers, staff, courses, fee rules, refund policy, discounts, RBAC, petty cash cap, leave policy, booking window, grace period, escalation thresholds, certificate master templates, social media approval, calendar push. |
| <b>Branch Admin</b>      | Branch           | Approves leave/early-out, petty cash (1st level), student emergency out, resource log oversight. Manages personalized classes, branch calendar, certificate issuance, social media drafts.                                     |
| <b>Teacher</b>           | Classes          | Geo-attendance, student attendance, homework, results, daily class update (mandatory). Requests leave. Views assigned class reports.                                                                                           |
| <b>Accountant</b>        | Finance          | Petty cash requests and receipts. Billing records. Cannot self-approve.                                                                                                                                                        |
| <b>Receptionist</b>      | Front desk       | Student check-in, appointment viewing, basic communication.                                                                                                                                                                    |
| <b>Janitor / Cleaner</b> | Maintenance      | Receives and closes maintenance tasks. Submits daily resource log.                                                                                                                                                             |
| <b>Student</b>           | Enrolled courses | Timetable, homework, results, invoices, calendar, certificates, announcements. Mobile app.                                                                                                                                     |
| <b>Parent / Guardian</b> | Linked students  | Student progress, invoices, attendance, calendar. Initiates communication. Books appointments. Applies student advance leave. Sees only assigned teachers contact info.                                                        |

## 5. Tenant & Branch Structure

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### 5.1 Hierarchy

|                              |                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Platform Level</b>        | Super Admin manages all tenants with full isolation between them.                                            |
| <b>Tenant Level</b>          | One institution. One PAN/VAT. Global policies for all branches.                                              |
| <b>Branch / Center Level</b> | Physical centers. Own classes, schedules, fee collections, staff. Centers and branches are the same concept. |
| <b>Class / Course Level</b>  | Regular (UKG–12), Music, Short-term, Long-term, Personalized.                                                |

### 5.2 Teacher Sharing

- Teachers assigned to a specific branch by default.
- Tenant Admin or Branch Admin can assign a teacher to multiple branches.
- Teacher data can be imported from one branch to another via admin panel.
- A teacher's geo-attendance radius is validated per branch scheduled that day.

## 6. Course & Class Structure

| Type                      | Enrollment                                                  | Structure                                                                                 | Fee Type                                                         |
|---------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Regular (UKG–12)</b>   | Via grade enrollment. Independent of other course types.    | Subjects customizable per center. Fixed weekly timetable, adjustable by admin.            | Monthly, term-based, or installment.                             |
| <b>Music</b>              | Separate enrollment. Concurrent with regular grade.         | Admin-defined. Schedule and curriculum independent.                                       | One-time, monthly, or installment.                               |
| <b>Short-Term Course</b>  | Separate enrollment. Concurrent with any enrollment.        | Admin-defined duration. Fixed end date.                                                   | One-time upfront or installment.                                 |
| <b>Long-Term Course</b>   | Separate enrollment. Concurrent with any enrollment.        | Base policy by Tenant/Super Admin. Admin customizes per course.                           | Monthly, installment, or term-based.                             |
| <b>Personalized Class</b> | Branch Admin creates for a specific student or small group. | One-on-one or small group. Custom timetable, subject, teacher. Outside regular timetable. | Per session/hour rate OR fixed package rate — defined per class. |

- A student can be simultaneously enrolled in a regular class + music + short/long-term + personalized class under the same tenant.
- Personalized classes follow separate fee, attendance, and homework rules defined at class creation.

## 7. Teacher Geo-Attendance System

### 7.1 Entry — Manual Mark IN

- Teacher taps "Mark IN" on mobile app. GPS validated against branch radius.
- Outside radius: Mark IN blocked. No override in Phase 1.
- Scheduled class at that time: counted as present for that session.
- No scheduled class: logged as "Unscheduled Presence" — not counted, visible in reports only.

### 7.2 Exit — Manual Mark OUT & Auto-Departure

- Teacher taps "Mark OUT" when leaving. Departure time recorded.
- Exits radius without tapping OUT: auto-departure fires.
- Auto-departure timestamp = last known GPS position inside radius.
- Auto-departure triggers immediate push notification to Branch Admin and is logged in reports.

### 7.3 Grace Period

- Tenant Admin defines grace period (minutes) per branch.
- Brief exit within grace window: no departure triggered.
- Grace period covers inter-session gaps between back-to-back classes.

### 7.4 Mid-Session Departure & Return

- Teacher auto-departed mid-class → returns → taps Mark IN → session resumes.
- New IN stamp created. Total in-premises time = sum of all valid IN→OUT intervals.
- All stamps (IN, OUT, AUTO-OUT, RE-IN) preserved in audit log.

### 7.5 Session Validation via Daily Class Update

- Correct IN/OUT does not fully confirm a session until daily class update is submitted.
- Missing update: session status = "Present — Update Pending".
- End-of-session cron flags pending updates and notifies teacher.

### 7.6 Session Status Reference

| Status                          | Condition                                               | Admin Notification                       |
|---------------------------------|---------------------------------------------------------|------------------------------------------|
| <b>Present (Confirmed)</b>      | IN + OUT + daily update submitted.                      | None.                                    |
| <b>Present — Update Pending</b> | IN + OUT correct, update missing.                       | Cron notifies teacher. Visible to admin. |
| <b>Partial Presence</b>         | Auto-departed mid-session. Time calculated from stamps. | Immediate notification to Branch Admin.  |

|                             |                                    |                                |
|-----------------------------|------------------------------------|--------------------------------|
| <b>Unscheduled Presence</b> | Marked IN with no scheduled class. | Reports only. No notification. |
| <b>Absent</b>               | No IN for a scheduled session.     | Daily attendance report.       |

## 7.7 Student Attendance

- Teacher marks each student present/absent from class roster in mobile app.
- Students with fee dues: flagged "Blocked" — cannot be marked present.
- Students with approved leave: pre-marked "Absent (Excused)" — no manual marking needed.

## 8. Fee & Billing

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### 8.1 Fee Structure

|                         |                                                                              |
|-------------------------|------------------------------------------------------------------------------|
| <b>Fee Definition</b>   | Per course/class by Branch Admin.                                            |
| <b>Fee Types</b>        | Monthly recurring, one-time, installment-based, term-based.                  |
| <b>Charge Start</b>     | From admission date, starting from exact class start date — not whole month. |
| <b>Discounts</b>        | Flat, %, sibling, scholarship. Applied by Accountant or Admin only.          |
| <b>Late Fee / Fines</b> | Default zero. Admin configures. Applied only after explicit config.          |
| <b>VAT / PAN</b>        | PAN at tenant level. Displayed on all invoices.                              |

### 8.2 Invoice Generation

- Combined invoice per billing cycle + separate per-course invoices.
- SMS: combined summary. In-app: itemized breakdown with dues, discounts, fines, net payable.
- One unique Nepal Pay QR per invoice per billing cycle.
- Auto-confirm via webhook. Manual fallback if webhook fails.

### 8.3 Fee Blocking & Refund

- Monthly cron (1st) checks dues → blocks students → teacher sees "Blocked" flag.
- Branch Admin can grant temporary access with mandatory logged reason.
- Refund policy (pro-rata, fixed deduction, no-refund) defined by Tenant Admin.

## 9. Leave & Early Out System

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### 9.1 Staff Leave

| Leave Type        | Approval Chain                 | Rules                                              |
|-------------------|--------------------------------|----------------------------------------------------|
| Casual Leave      | Branch Admin approves/rejects. | Line manager is always Branch Admin.               |
| Early Out Request | Branch Admin approves/rejects. | Same flow as casual leave.                         |
| Long Sick Leave   | Branch Admin → Tenant Admin.   | Branch approves first, then Tenant Admin confirms. |

### 9.2 Student Leave

- Advance (planned): parent applies → Branch Admin + teachers notified → approved → auto-excused in attendance.
- Emergency out: Branch Admin logs → immediate push to parents → logged in student record.

## 10. Appointment Booking System

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|                                |                                                                              |
|--------------------------------|------------------------------------------------------------------------------|
| <b>Group Meeting</b>           | One common slot. All requested parties must approve.                         |
| <b>Individual Appointments</b> | Separate slot per teacher/admin. Each handled independently.                 |
| <b>Booking Window</b>          | Minimum 24 hours advance. Configurable by Tenant Admin.                      |
| <b>Alternative Proposal</b>    | Teacher/admin can propose alternative time. Becomes independent negotiation. |
| <b>Privacy</b>                 | Parent can only book with teachers assigned to their child.                  |
| <b>Notifications</b>           | Push + SMS for: request, approval/rejection/alternative, confirmation.       |
| <b>Phase 1</b>                 | In-person only. Virtual meeting links planned for future.                    |

## 11. Petty Cash Management

| Step                      | Actor               | Action / Notes                                                |
|---------------------------|---------------------|---------------------------------------------------------------|
| <b>1 — Request</b>        | Accountant          | Submits purpose, amount. Cannot exceed remaining monthly cap. |
| <b>2 — Level 1 Review</b> | Branch Admin        | Approves or rejects with reason.                              |
| <b>3 — Level 2 Review</b> | Tenant Admin        | Approves, rejects, or sends back for revision with remarks.   |
| <b>4 — Release</b>        | System              | Auto-releases. Notifies accountant. Updates cap balance.      |
| <b>5 — Receipt</b>        | Accountant          | Submits proof. Open-ended — no deadline.                      |
| <b>6 — Close</b>          | Branch/Tenant Admin | Verifies proof and manually closes the record.                |

- Monthly cap set by Tenant Admin. Resets 1st of every month.
- Each record tracks: purpose, amount, approval chain + timestamps, release date, receipt status, closing date.

## 12. Resource & Infrastructure Logging

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- Daily mandatory log per classroom (marker, duster, board, room condition, remarks, action required).
- Must be completed before staff can proceed — mandatory gate like teacher daily update.
- Specific staff role pre-configured per classroom/resource type by Tenant Admin.
- Action Required = Yes: system auto-assigns task to pre-configured staff. Branch Admin can override.
- Task completion: staff marks done → timestamp + responsible staff member auto-logged.
- Escalation: unresolved tasks beyond admin-defined X days → notification to Branch Admin / Tenant Admin.

## 13. Communication Hub

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|                                 |                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------|
| <b>Push Notifications</b>       | Real-time alerts for all app users.                                                       |
| <b>SMS</b>                      | Parents/guardians. Fee alerts, attendance summaries, urgent notifications.                |
| <b>Institutional Broadcasts</b> | Admin sends announcements to all or specific groups. One-way.                             |
| <b>Threaded Messaging</b>       | Parent initiates to specific teacher/admin. Threaded per student. Full history preserved. |
| <b>System Notifications</b>     | Auto-generated: fee due, attendance, homework, leave, appointment, emergency out.         |
| <b>Privacy</b>                  | Parents see only assigned teachers. Teachers see only their students' parents.            |

## 14. ERP & Accounting

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|                              |                                                                        |
|------------------------------|------------------------------------------------------------------------|
| <b>Center-Level Expenses</b> | Rent, utilities, maintenance, local staff costs.                       |
| <b>Tenant-Level Expenses</b> | Platform subscriptions, shared/corporate costs.                        |
| <b>Petty Cash</b>            | Two-level approval. See Section 11.                                    |
| <b>Payroll</b>               | Fixed monthly OR per-class/hour auto-calc from timetable + attendance. |
| <b>Monthly P&amp;L</b>       | Per branch + consolidated across all branches.                         |
| <b>Ledger Export</b>         | Double-entry CSV/PDF for manual accountant use.                        |

## 15. Automation & Cron Jobs

| Cron Job                     | Schedule                | Description                                                                     |
|------------------------------|-------------------------|---------------------------------------------------------------------------------|
| Monthly Due Verification     | 1st of month            | Checks unpaid dues. Auto-blocks students. Notifies admin.                       |
| Fee Reminder SMS             | Day 1 + Day 3 + custom  | Payment reminders to parents per tenant-defined schedule.                       |
| Homework Auto-Distribution   | On creation + scheduled | Distributes homework to enrolled students. Recurring templates run on schedule. |
| Test / Exam Reminders        | Per schedule            | Push + SMS to students and parents before upcoming tests.                       |
| Salary Reminder              | 25th of month           | Notifies admin. Auto-calculates per-class/hour salaries.                        |
| Attendance Report Generation | Daily + Weekly          | Summaries to admin and parents.                                                 |
| Teacher Session Flag         | End of each session     | Flags missing daily class updates. Notifies teacher.                            |
| Resource Task Escalation     | Daily check             | Escalates unresolved tasks beyond X days.                                       |
| Petty Cash Cap Reset         | 1st of month            | Resets monthly petty cash usage counter per tenant.                             |
| Contract Expiry Alerts       | Daily check             | Flags staff documents/contracts expiring within 30 days.                        |
| Performance Score Update     | Weekly                  | Recalculates staff and student performance scores from latest data.             |
| Financial Forecast Update    | Monthly                 | Refreshes enrollment-based fee forecast and AI expense suggestions.             |

## 16. Technical Architecture

| Layer                   | Technology                                                                               |
|-------------------------|------------------------------------------------------------------------------------------|
| <b>Web Frontend</b>     | React.js. Role-scoped dashboards for all roles.                                          |
| <b>Mobile App</b>       | React Native (iOS + Android). Geo-attendance, student/parent portal, push notifications. |
| <b>Backend</b>          | Node.js RESTful API. Modular service architecture. JWT authentication.                   |
| <b>Database</b>         | PostgreSQL with row-level tenant isolation. All queries scoped by tenant_id.             |
| <b>Payments</b>         | Nepal Pay. Per-invoice QR. Webhook auto-confirm.                                         |
| <b>Notifications</b>    | Push: Firebase Cloud Messaging (FCM). SMS: Nepal SMS gateway.                            |
| <b>File Storage</b>     | Cloud object storage for documents, homework, receipts, certificates, photos.            |
| <b>Geofencing</b>       | Mobile GPS + server-side radius validation. Grace period managed server-side.            |
| <b>Job Scheduling</b>   | Node-cron or queue-based system for reliable job execution.                              |
| <b>Export Engine</b>    | PDF generation (Puppeteer/PDFKit) + Excel (ExcelJS) for reports, certificates, ledger.   |
| <b>AI / ML</b>          | Forecasting engine for financial estimation and expense suggestions (Phase 2).           |
| <b>Social Media API</b> | Meta Graph API (Facebook/Instagram), TikTok API, LinkedIn API for post scheduling.       |

## 17. Non-Functional Requirements

| Requirement      | Specification                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------|
| Availability     | 99.5% uptime SLA. Scheduled maintenance during off-peak hours.                                           |
| Performance      | Page load < 2s on 4G. API response < 500ms for standard queries.                                         |
| Security         | JWT + refresh tokens. HTTPS only. Role-based data access at API layer. Sensitive data encrypted at rest. |
| Data Isolation   | Row-level tenant isolation in PostgreSQL. No cross-tenant data leakage. Super Admin access auditable.    |
| Mobile           | React Native. iOS 13+, Android 8+. GPS required for teacher attendance.                                  |
| Scalability      | Horizontal scaling of Node.js. DB connection pooling. Per-tenant cron isolation.                         |
| Compliance       | VAT/PAN on all invoices per Nepal tax law. Nepal Pay compliance. Social media API ToS compliance.        |
| Offline Handling | Clear error state when offline. No offline attendance submission.                                        |
| Audit Logging    | All admin actions logged with timestamp and actor.                                                       |
| Exportability    | All major reports exportable as PDF and Excel. Ledger as CSV/PDF.                                        |

## 18. Social Media Integration

TMS provides an integrated social media post scheduling and management tool, allowing institutions to manage their online presence directly from the platform without switching to external tools.

### 18.1 Supported Platforms

- Facebook (via Meta Graph API)
- Instagram (via Meta Graph API)
- TikTok (via TikTok API)
- LinkedIn (via LinkedIn API)

### 18.2 Post Management Flow

1. Branch Admin creates a post: text + images/videos, selects target platform(s), sets scheduled publish date/time.
2. Post is submitted for review — status: "Pending Approval".
3. Tenant Admin receives notification. Reviews the post in the approval queue.
4. Tenant Admin approves → post is queued for scheduled publish. Or rejects with remarks → Branch Admin is notified to revise.
5. At scheduled time, TMS publishes to connected platform(s) via API.
6. Post status updates to "Published". Publish timestamp logged.

### 18.3 Access Rules

|                     |                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| <b>Branch Admin</b> | Can create, edit, and delete draft posts. Can view published posts. Cannot publish directly.      |
| <b>Tenant Admin</b> | Can create, approve, reject, and publish posts. Manages platform API credentials at tenant level. |
| <b>Super Admin</b>  | Platform-level API key management. Cannot access tenant social content.                           |

### 18.4 Content Rules

- Supported content: text, images, and videos. Character limits enforced per platform.
- Failed publishes (API errors) are logged and admin is notified for manual retry.
- All post history (drafted, approved, rejected, published) is stored and auditable.
- Platform API credentials (OAuth tokens) stored securely at tenant level.

## 19. Certificate Generation System

TMS generates and issues certificates as PDF documents directly from the platform. Certificates are manually issued by admin. No auto-trigger on course completion.

### 19.1 Certificate Types

- Course Completion Certificate — issued on completion of any course type.
- Achievement / Merit Certificate — issued for academic or extracurricular achievement.
- Attendance Certificate — issued based on attendance percentage for a period.
- Custom Certificate — any admin-defined purpose using a custom template.

### 19.2 Template Management

|                     |                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Tenant Admin</b> | Creates and manages master certificate templates for the institution. Defines layout, branding, fields, and content blocks.    |
| <b>Branch Admin</b> | Customizes branch-specific details (branch name, address, logo) on top of the master template. Cannot change master structure. |

### 19.3 Issuance Rules

- Certificates are manually issued by Branch Admin or Tenant Admin only.
- On issuance, a PDF is generated and stored against the student record.
- Student and parent receive a push notification and can download the certificate from the app.
- Each certificate is assigned a unique certificate ID for verification purposes.
- Certificate issuance history is logged per student with issuer, date, and type.

## 20. Academic Calendar & Payment Calendar

### 20.1 Academic Calendar

The academic calendar provides a full-year view of scheduled events, holidays, exam dates, and course milestones.

|                              |                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tenant Admin Calendar</b> | Defines institution-wide events: public holidays, exam periods, annual ceremonies, fee due dates. Pushed to all branches.                           |
| <b>Branch Admin Calendar</b> | Can add branch-specific events on top of the tenant calendar (e.g., local holiday, branch test date). Cannot modify or remove tenant-pushed events. |
| <b>Visibility</b>            | Full calendar visible to students and parents in the mobile app.                                                                                    |
| <b>Upcoming Events</b>       | Quarterly view of upcoming events displayed on dashboard for admin, teachers, students, and parents.                                                |

### 20.2 Payment Calendar

An enhancement of the existing billing module — provides a calendar-based view of upcoming payment due dates.

- Displays all upcoming fee due dates for enrolled courses in a monthly/weekly calendar view.
- Visible to: admin (all students), teachers (their class students), parents (their child).
- Color-coded: upcoming (green), due soon within 3 days (amber), overdue (red).
- Clicking a due date shows invoice details and Nepal Pay QR for that billing cycle.

## 21. Personalized Classes

Personalized classes are one-on-one or small group sessions created specifically for a student or a defined group, outside the regular class timetable.

### 21.1 Creation & Enrollment

- Created by Branch Admin only.
- Admin defines: student(s), assigned teacher, subject/topic, schedule (day/time), duration, and fee structure.
- Enrollment is separate from all regular and course enrollments.

### 21.2 Fee Rules

|                                |                                                                               |
|--------------------------------|-------------------------------------------------------------------------------|
| <b>Per Session / Hour Rate</b> | Fee calculated based on actual sessions attended × rate.                      |
| <b>Fixed Package Rate</b>      | One-time or recurring fixed fee for a defined number of sessions or duration. |
| <b>Billing</b>                 | Follows same invoice, Nepal Pay QR, and blocking rules as other courses.      |

### 21.3 Attendance & Homework

- Teacher takes attendance per session, same as regular classes.
- Homework can be assigned and is specific to the personalized class enrollment — not shared with regular class roster.
- Geo-attendance rules apply to the assigned teacher for personalized class sessions.

### 21.4 Reporting

- Progress tracked separately from regular class performance.
- Admin and parent can view session history, attendance, and homework completion for personalized classes.

## 22. Student Performance Tracking

TMS tracks student academic performance across time and against class peers, providing upgrade/downgrade signals to teachers, admins, and parents.

### 22.1 Performance Metrics

|                                   |                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Test Scores</b>                | Weekly and periodic test results per subject per course.                                                    |
| <b>Trend Analysis</b>             | Week-over-week and month-over-month performance change per subject and overall.                             |
| <b>Class Comparison</b>           | Student score vs class average per subject. Shown as above/at/below average.                                |
| <b>Upgrade / Downgrade Signal</b> | Automatically flagged when performance improves or declines beyond a configurable threshold (set by admin). |
| <b>Strong / Weak Subjects</b>     | Identified from average scores across multiple tests.                                                       |
| <b>Attendance Impact</b>          | Correlation between attendance rate and performance scores shown in analytics.                              |
| <b>Administrative Remarks</b>     | Branch Admin can add remarks per student visible to admin and teachers. Configurable visibility to parents. |
| <b>Teacher Remarks</b>            | Teacher can add per-student remarks per class session. Visible to admin and parents.                        |
| <b>Active Students Count</b>      | Dashboard widget: number of currently active students per branch, per class, per course.                    |

### 22.2 Visibility

- Admin: full view across all students and all metrics.
- Teacher: view limited to students in their assigned classes.
- Parent: view limited to their linked child. Upgrade/downgrade signals, trends, and remarks visible.
- Student: view their own scores, trends, and strong/weak subjects.

## 23. Financial Estimation & AI Suggestions

Phase 2 introduces an intelligent financial layer that forecasts expected revenue, flags missing expense entries, and provides AI-driven suggestions to help admins plan institutional finances.

### 23.1 Monthly Financial Forecast

|                            |                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Base Forecast</b>       | Calculates expected monthly fee income from current enrollments × applicable fee rates per course per student.               |
| <b>Seasonal Trends</b>     | Factors in historical enrollment patterns (peak months, low months) to adjust forecast. Uses prior-year data once available. |
| <b>Dropout Adjustment</b>  | Applies estimated attrition rate (configurable by admin or derived from historical data) to reduce forecast.                 |
| <b>Actuals vs Forecast</b> | Dashboard compares actual fee collected vs forecast for the month. Variance highlighted.                                     |

### 23.2 AI-Driven Expense Suggestions

- Rule-based alerts: "Last month rent was recorded. No rent entry yet for this month."
- Pattern-based suggestions: "Utility expense is typically logged by 5th of the month. None recorded yet."
- Anomaly detection: "This month salary total is 30% higher than last 3-month average. Please verify."
- Budget planning prompts: "Based on current enrollment, projected income is NPR X. Estimated fixed costs are Y. Surplus/deficit: Z."

### 23.3 Yearly Financial Tracking

- Full-year income vs expense view per branch and consolidated across tenant.
- Month-by-month breakdown: fee collections, salary, petty cash, rent, utilities, other expenses.
- Exportable as PDF or Excel for accountant or management review.
- Year-on-year comparison once multi-year data is available.

## 24. HR Management

TMS includes a comprehensive HR module covering staff records, document management, performance appraisals, and formal exit processes.

### 24.1 Staff Records

|                               |                                                                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Information</b>   | Name, contact, address, emergency contact, date of birth, NID/passport.                                                              |
| <b>Employment Details</b>     | Joining date, designation, department, branch assignment, contract type, salary structure.                                           |
| <b>Document Uploads</b>       | Employment contract, NID copy, academic certificates, professional qualifications, photo ID. Each document has an expiry date field. |
| <b>Document Expiry Alerts</b> | Daily cron checks expiry dates. Alerts sent to Branch Admin and Tenant Admin 30 days before expiry.                                  |
| <b>Leave Balance</b>          | Tracks earned, used, and remaining leave days per staff member per leave type.                                                       |

### 24.2 Staff Performance Score

Each staff member has a continuously updated performance score calculated from the following components:

| Component                                  | Weight         | Source                                                     |
|--------------------------------------------|----------------|------------------------------------------------------------|
| <b>Attendance Rate</b>                     | Configurable % | Geo-attendance records vs scheduled sessions.              |
| <b>Daily Class Update Compliance</b>       | Configurable % | Ratio of sessions with completed update vs total sessions. |
| <b>Student Feedback Rating</b>             | Configurable % | Average rating given by students at end of term/course.    |
| <b>Parent Feedback Rating</b>              | Configurable % | Average rating given by parents via app.                   |
| <b>Leave Compliance</b>                    | Configurable % | Proper leave application vs unplanned absences.            |
| <b>Task Completion (Maintenance Staff)</b> | Configurable % | Ratio of assigned tasks completed within deadline.         |

- Component weights are configurable by Tenant Admin. Must total 100%.
- Score is recalculated weekly by a cron job.
- Score history is stored — admin can view trend over time.
- Tenant Admin and Branch Admin can view all staff scores. Teachers can view their own score only.

## 24.3 Formal Exit Flow

7. Staff member or Branch Admin initiates resignation in system. Resignation date and reason recorded.
8. Notice period is calculated based on contract terms stored in HR record.
9. Clearance checklist is generated: return of assets, handover of classes/tasks, pending task resolution.
10. Final salary is auto-calculated: pro-rated salary for the month + any pending allowances - any dues.
11. Branch Admin reviews and confirms clearance checklist completion.
12. Tenant Admin approves final settlement.
13. Staff account is deactivated. Record archived — not deleted. Historical data preserved.

## 24.4 HR Reports

- Active staff count per branch per role.
- Staff attendance and leave summary per month.
- Performance score leaderboard per branch.
- Upcoming contract/document expiries.
- Exit history with dates, reasons, and settlement amounts.

## 25. Phased Build Plan (Updated v2.0)

| Phase   | Focus                               | Deliverables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | Core Foundation                     | Tenant/branch setup · RBAC · Student & teacher enrollment · Regular class timetable · Full geo-attendance engine · Fee billing · Nepal Pay QR & webhook · VAT/PAN invoicing · Student blocking · Smart dashboard · Digital student ID · Leave & early out · Student emergency departure · Push + SMS foundation                                                                                                                                                                                                                                                                               |
| Phase 2 | Academic, Operations & Intelligence | Homework + auto-distribution · Results & performance · Teacher verification · Communication hub · Report export · Teacher rating & feedback · Appointment booking · Resource logging · Student knowledge graph · Certificate generation · Academic & payment calendar · Social media scheduling & approval · Personalized classes · Student performance tracking (trends + class avg) · HR management (records, documents, expiry alerts, performance scoring, exit flow) · AI financial estimation & expense suggestions · Yearly financial tracking · Canteen management · Vehicle tracking |
| Phase 3 | ERP & Advanced                      | Full expense management · Payroll auto-calculation · Petty cash approval engine · P&L dashboard · Double-entry ledger export · Music class module · Short/long-term course engine · Refund management · Full cron automation suite · Task escalation engine · Performance analytics deep-dive                                                                                                                                                                                                                                                                                                 |

## 26. Open Items for Next Phase

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- Database schema design (ERD) — multi-tenant tables, RBAC, attendance stamps, billing, HR, social media, certificates.
- API contract design — endpoint definitions, authentication middleware, request/response shapes.
- Nepal Pay integration spec — QR generation flow, webhook payload, error handling.
- Social media API OAuth flow — token storage, refresh handling, rate limits per platform.
- SMS gateway vendor selection — rate limits, fallback handling.
- Mobile GPS accuracy thresholds — minimum accuracy required for valid IN stamp.
- AI/ML forecasting model design — data inputs, model type, retraining schedule.
- Certificate PDF template engine — template schema, field mapping, rendering library.
- Tenant onboarding flow — signup, first branch config, go-live checklist.
- Super Admin billing/subscription module — how tenants are billed for platform usage.
- Data migration plan — for institutions moving from manual/spreadsheet systems.
- Canteen system design — cashless wallet architecture, POS integration options.
- Vehicle tracking system design — GPS device vs driver mobile, parent visibility scope.